Experiment 11

Aim

To develop an Android application that allows users to enter data and store it in the device's external storage (SD card).

Procedure

1. Open Android Studio and create a new project using **Empty Views Activity**.
2. Design the user interface using:
    - **EditText** for user input
    - **Button** to save data
3. Add required permission in AndroidManifest.xml:
    - `WRITE_EXTERNAL_STORAGE`
4. Open `MainActivity.java`.
5. Link UI components using findViewById().
6. Use **MediaStore API** to save data into external storage (Downloads folder).
7. Handle empty input cases.
8. Display confirmation using **Toast message**.
9. Run the application and verify file creation.

Code

XML Layout (activity_main.xml)

```XML
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/inputText"
        android:hint="Enter data"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/saveBtn"
        android:text="Save to SD Card"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

Java Code (MainActivity.java)

```java
package com.example.experiment_11;


import android.content.ContentValues;
import android.os.Bundle;
import android.os.Environment;
import android.provider.MediaStore;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.io.OutputStream;

public class MainActivity extends AppCompatActivity{

    EditText inputText;
    Button saveBtn;

    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        inputText = findViewById(R.id.inputText);
        saveBtn = findViewById(R.id.saveBtn);

        saveBtn.setOnClickListener(v -> {
            String data = inputText.getText().toString();

            if (data.isEmpty()) {
                Toast.makeText(this, "Enter some data",
Toast.LENGTH_SHORT).show();
            } else {
                saveToDownloads(data);
            }
        });
    }

    private void saveToDownloads(String data) {
        try {
            String fileName = "mydata_" +
System.currentTimeMillis() + ".txt";

            ContentValues values = new ContentValues();
            values.put(MediaStore.MediaColumns.DISPLAY_NAME,
fileName);
            values.put(MediaStore.MediaColumns.MIME_TYPE,
"text/plain");
            values.put(MediaStore.MediaColumns.RELATIVE_PATH,
Environment.DIRECTORY_DOWNLOADS);

            var resolver = getContentResolver();
            var uri =
resolver.insert(MediaStore.Files.getContentUri("external"),
values);
```

```
            if(uri != null) {
                OutputStream os = resolver.openOutputStream(uri);
                os.write(data.getBytes());
                os.close();

                Toast.makeText(this, "Saved to Downloads!",
Toast.LENGTH_LONG).show();
            }else {
                Toast.makeText(this, "Failed to save file",
Toast.LENGTH_SHORT).show();
            }

        }catch(Exception e) {
            Toast.makeText(this, "Error saving file",
Toast.LENGTH_SHORT).show();
            e.printStackTrace();
        }
    }
}
```

Output

Result

Thus, an Android application was successfully developed to store user input data in the device's external storage (SD card).